PLANT DISEASE CLASSIFICATION

About project

Project is about to create a deep learning system for plant disease classification.The goal is to predict a disease type,regardless of the plant species.Dataset consists of 8000 images and 40 disease types.It has severe class imbalance 32:1.The biggest class has 840 images the smallest only 26.Dataset was divided into train and validation folders,and in validation one class was missing so I moved a small number of samples from train to val,but this caused even more imbalance 40:1. So a better approach would be to collect more samples from internet,and not only for this class but for all which had smaller data.

Methodology

Two approaches were considered:first is to design architecture from scratch,but this was not an optimal solution, because it requires a large dataset and careful architecture design.Second is transfer learning, were we take pretrained models and fine-tune for the task.So I choose second approach.Testing different architectures like EfficientNet-B0, EfficientNet-B3, ResNet18, ResNet50 best results I got with ResNet50.But I do not think that problem was architecture design,it was preprocessing and some rare classes which were dropping scores.The original classification head in ResNet50 was replaced with a custom architecture.

```
model.fc = nn.Sequential(
nn.Dropout(p=0.4),
nn.Linear(in_features, 512),
nn.ReLU(),
nn.Dropout(p=0.3),
nn.Linear(512, num_classes)
)
```

Training strategy

I used two stage training startegy:In first stage models backbone was frozen and only classification head was traing 10 epochs with 1e-4 learning rate. Second stage was fine-tuning,all layers were unfrozen and entire was traing 30 epochs with learning rate 1e-3.Additionaly I used Cosine Annealing scheduler for both stages to gradually reduce the learning rate and early stopping was used to prevent overfitting. The best model was saved based on validation mAP score.

Handling class imbalance

As dataset is imbalanced 40:1 I applied some techniques,which helped to get better results,but not the best.First technique is WeightedRandomSampler with square-root balancing to oversample rare classes second is focal loss instead of standard cross entropy.Focal loss reduces the weights of big classes and focusees training on hard rare samples.

Data augmentation

As for plant diseases color is an important feature heavy augmentation led to worse

performance.Unfortunatly I did not save first augmentation strategies,but this is the final one,with slightly color changing,rotation and flipping.

```
transforms.Resize((256, 256)),
transforms.RandomResizedCrop(224, scale=(0.85, 1.0)),
transforms.RandomHorizontalFlip(p=0.5),
transforms.RandomRotation(15),
transforms.ColorJitter(
brightness=0.15,
contrast=0.15,
saturation=0.1,
hue=0.01
),
transforms.ToTensor(),
transforms.Normalize([0.485, 0.456, 0.406],
[0.229, 0.224, 0.225])
```

I also used Mixup augmentation with alpha = 0.2,which slightly improved results

Ablations

I tested several combinations and techniques.
First is model architecture: EfficientNet-B0, EfficientNet-B3, ResNet18, ResNet50 got best results with last one.
For loss function classic CrossEntropy and Focal loss,Focal loss gave better results for rare classes
Augmentation: Aggressive and light
With/without sampler: Sampler improved batch balance significantly
Different MixUp alpha values: 0.2 was more stable
Extra tarining for hard classes(no results)
Different learning rates
Weighted CrossEntropy caused double penalization with sampler, dropped it
Different batch size 32 and 64

Final results

```
LR:           1.74e-05
Train Loss:    0.5555
Val Loss:      0.4687
mAP:          0.8645
F1 Macro:      0.7889
F1 Weighted:   0.7876
Precision:     0.8085
Recall:       0.7900
Mean cls acc:  0.7900
Worst cls acc: 0.2000 (class 31)

Worst 3 AP:    [(33, '0.40'), (31, '0.44'), (1, '0.59')]
```

Three classes 31,33 and 2 showed the worst results almost all training time, I tried to train model more 10 epoches only with this worst classes,but did not get any results,so because of

them all scores are lower.Fr making better, I think that we can add more samples for this classes and spots should be clearly visible.

Conclusion

Transfer learning is effective for plant disease classification even with small and imbalanced dataset.For plants it is important not to use hard augmentation,because color is their weak part.Class imbalance impacts on accuracy and other metrics and make them lower.For future improvements I would collect more data for rare classes, and make sure that images have good quality and spots are clearly visible,also try more models and write one on my own.

The model is deployed as a REST API with Swagger documentation:
API: https://arpigmm-plant-disease-api.hf.space
Docs: https://arpigmm-plant-disease-api.hf.space/docs
Model weights: https://huggingface.co/arpigmm/plant-disease-model